The result shows that the temperature at which the carbonate was formed varied during the life of the creature; further, the creature that inhabited the shell must have died during the spring of its fourth year since there was no further growth of the shell after the forth period of increasing temperature. Thus, by using this method it is possible to determine a small window of yearly temperature variations millions of years back in history.

Chemical isotope effects are particularly large for lighter elements in biological systems. The *chlorella algae* prefers deuterium over hydrogen, and tritium over deuterium. The enrichment factor depends on the conditions of growth; for deuterium to hydrogen an enrichment value of $1.6 - 3$ has been found, while for tritium to hydrogen the enrichment factor is about 2.5. Bacteria behave similarly, e.g. *coli bacteria* showing an enrichment factor for deuterium of 3.9.

In as much as some of the hydrogen atoms are not exchanged readily due to the inertness of their chemical bonds, the isotopic fractionation which involves the easily exchangeable hydrogen atoms in these biological processes must have even larger enrichment factors for deuterium and tritium than their measured values would indicate.

The peculiarity of biological material to prefer certain isotopes has led to studies of how biological material behaves in an isotopic environment which differs substantially from that found in nature. Normally it is found that the organisms wither away and lose their ability to reproduce. Carp cannot survive a higher D_2O concentration than 30%, but, on the other hand, some organisms show a strong growth, and some microorganisms have been found to be able to live in pure D_2O or $H_2{}^{18}O$. It has been possible to raise mice with only ^{13}C in their organism (fed on ^{13}C algae). Exchanging natural $^{14}NH_3$ for $^{15}NH_3$ seems to have little effect on biological systems.

In all of these investigations it should be noted that even when we characterize an isotopic effect as large, it is still quite small by normal reaction criteria except for hydrogen isotopes. As pointed out previously, we can assume that for all but the very lightest elements in most chemical experiments that there is no isotope effect. This assumption forms a basis of the use of radioactive tracers to study chemical systems and diffusion at constant concentration.

3.7. ISOTOPE EFFECTS IN CHEMICAL KINETICS

The reason why higher organisms cannot survive when all light hydrogen atoms are replaced by deuterium is to be found not so much in a shift of chemical equilibria as in a shift in reaction rate leading to a fatal lowering of the metabolic rate when light isotopes are replaced by heavier.

In contrast to chemical equilibria, chemical reaction rates depend on the concentration of the reactants, the transition states and the activation energy but not on the

product. The concentration of the transition states depends on the activation energy for its formation and the frequency for its decomposition into the products. These factors can be derived from the partition function which, as mentioned above, differ slightly for molecules of different isotopic composition. Let us consider the reaction

$$A + BC \rightarrow AB + C \tag{3.49}$$

The rate constant is given by the expression

$$-d[A]/dt = k[A][BC] \tag{3.50}$$

The reaction is assumed to take place over an intermediate compound ABC, usually denoted $ABC^{\#}$ where the # indicates a short-lived transition state. According to the transition state theory, derived by H. Eyring, J. Biegeleisen, and others, it is assumed that the intermediate complex undergoes internal vibrations, with such an energy E_v that the bond is broken along the vertical line in the complex AB|C, leading to the fragments AB and C. The rate of reaction is the rate at which the complex $ABC^{\#}$ decomposes into the products. It can therefore also be written

$$-d[A]/dt = \nu\left[ABC^{\#}\right] \tag{3.51}$$

where ν is the frequency at which the complex decomposes.

The reaction can be schematically depicted as in Figure 3.7, where indices 1 and 2 refer to two isotopic reactant molecules (e.g. H_2O and HDO), which must have different zero point energies with frequencies ν_1 and ν_2, respectively. For simplicity only the

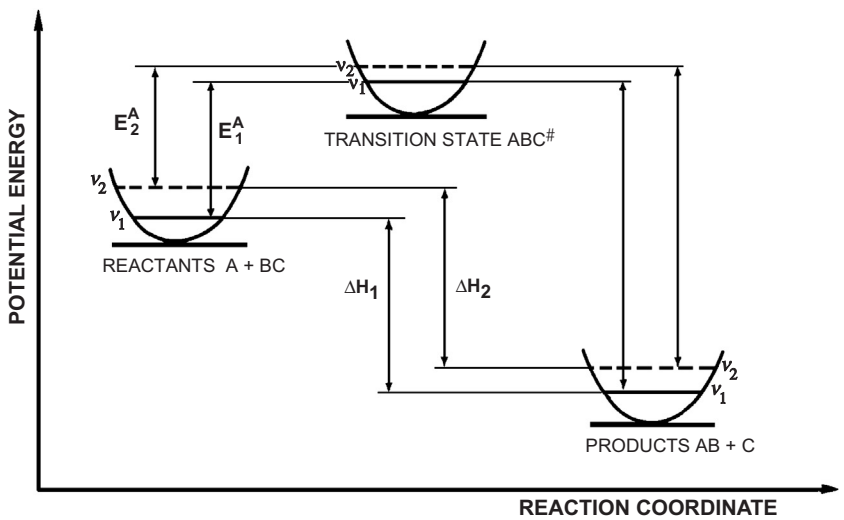

Figure 3.7 Schematic description of energies in a bimolecular reaction.

vibrational ground state is indicated; thus the energy change when going from reactants to products corresponds to the heat of reaction at absolute zero, ΔH_1 (0 K) and ΔH_2 (0 K) respectively. Because of the lower vibrational energy of the molecule indexed 1, this must contain the heavier isotope. In general, the difference in activation energy (E^{A}) is greater than the difference in heat of reaction ΔH for isotope molecules; thus, generally, isotope effects are larger in the kinetic than in the equilibrium effects.

When the molecule $\mathrm{ABC}^{\#}$ decomposes into AB and C, the vibrational energy, given by the Planck relation

$$E_{\nu} = \mathbf{h}\nu \tag{3.52}$$

is changed into kinetic energy of the fragments, whose energy is (see §3.6.2)

$$E_{\mathrm{kin}} = \mathbf{k}T \tag{3.53}$$

Because of the law of conservation of energy

$$E_{\nu} = E_{\mathrm{kin}} \tag{3.54}$$

and

$$\mathbf{h}\nu = \mathbf{k}T \tag{3.55}$$

This development assumes that the vibrational energy is completely converted to fragment translational energy. This assumption is not always valid for polyatomic fragments, in which there is also contribution from vibration and rotation energies. Introducing (3.55) into (3.51) and equating (3.50) and (3.51) yields

$$k[\mathrm{A}][\mathrm{BC}] = \mathbf{k}T\mathbf{h}^{-1}\left[\mathrm{ABC}^{\#}\right] \tag{3.56}$$

It is assumed that $\mathrm{ABC}^{\#}$ is in dynamic equilibrium with the reactants A and BC. Thus

$$\left[\mathrm{ABC}^{\#}\right]/([\mathrm{A}][\mathrm{BC}]) = k^{\#} \tag{3.57}$$

According to (3.27)

$$k^{\#} = F_{\mathrm{ABC}^{\#}}/(F_{\mathrm{A}}\, F_{\mathrm{BC}}) \tag{3.58}$$

which with (3.56) yields

$$k = \mathbf{k}T\, F_{\mathrm{ABC}^{\#}}/(\mathbf{h}\, F_{\mathrm{A}}\, F_{\mathrm{BC}}) \tag{3.59}$$

This expression must be multiplied by a factor κ, which is the probability that the complex will dissociate into products instead of back into the reactants as assumed in (3.57). The factor κ is called the *transmission coefficient*. The final rate expression thus becomes:

$$k = \kappa\, \mathbf{k}\, T\, F_{\mathrm{ABC}^{\#}}/(\mathbf{h}\, F_{\mathrm{A}}\, F_{\mathrm{BC}}) \tag{3.60}$$

As is shown in §3.6, the grand partition functions F_i can be calculated from theory and spectroscopic data; because these functions are mass dependent k is mass dependent. In calculating the F_i's, all modes of energy must be included as well as the population of the different energy states.

For isotopes of the lighter elements, the activation energy term makes the main contribution to the reaction rate isotope effect, while for the heavier elements the vibrational frequency causing the decomposition into the products plays the larger role. Because the energy states usually are more separated for the isotopic molecules of the products and reactants than for the transition state, isotope effects are usually larger in reaction kinetics than in equilibria.

Studies of kinetic isotope effects are of considerable theoretical interest, particularly in organic chemistry. The practical applications are still meagre, but this will not necessarily be so in the future. An example is the decrease in metabolic rate for ^{13}C compounds, which has led to the suggestion of its use for treatment of certain diseases, as e.g. porphyria.

3.8. ISOTOPE SEPARATION PROCESSES

Many fields of fundamental science have found great advantage in using pure or enriched isotopes. The industrial use of nuclear power also requires the enrichment of particular isotopes, primarily of the uranium fuel but also separation of H, D and T in heavy water moderated reactors as well as nitrogen isotope separation for nitride fuels to be used in modern fast reactors (Ch. 21). The methods which have been developed to achieve isotopic fractionation may be divided into two groups.

(a) *Equilibrium processes* (§3.6). These processes consume little energy, but the size of the isotope effect in normal chemical equilibrium limits their use to the isotope fractionation of very light elements, usually atomic number less than about 10.

(b) *Rate processes* (§3.7). This includes processes which depend on such phenomena as ionic mobility, diffusion, electrolysis, electromagnetic separations, centrifugation, and kinetic processes of a chemical nature. While the isotopic effects in these processes are normally larger than for equilibrium processes, they require a large amount of energy and therefore have economic restrictions.

3.8.1. Multistage processes

Figure 3.8 shows a flow scheme for an isotopic fractionation process (or isotope *enrichment* process) based on an α-value near 1. Each *stage* consists of a number of *cells* coupled in parallel; in the Figure only one cell is shown for each separation stage, but in order to obtain a high product flow, the number of cells are usually high at the feed point and then decrease towards the product and waste stream ends. Each cell contains a physical arrangement, which leads to an isotope fractionation. Thus the atomic fraction